

Your Company is a willing buyer of such produce, whilst the growers are free to sell to the highest bidder in the open market. Today, this programme covers over 80,000 hectares and has provided over 35 million mandays of employment. This approach has also brought with it a multiplicity of additional benefits – creation of a green cover for carbon sequestration, ground water recharge, regeneration of biomass and the nurturing of depleted soils. The green cover created has also enabled your Company to achieve a unique distinction of being a carbon positive company for 3 consecutive years.



It is immensely satisfying that your Company's conscious strategy to shed reliance on cheaper imports and instead invest in longer term benefits of promoting plantations, has yielded so many positive results. A business which was nearly sick a decade ago, is today not only a world-class producer of environmentally friendly, state-of-the-art elemental chlorine free paper & paperboard – the only one of its kind in India – but is also a significant contributor to the larger cause of sustainable livelihoods and environmental security.

Multiple Applications of Agro-Forestry

The paper and pulp industry, however, is only one application of wood-based industries, and barely uses 4% of the total wood consumed in India. Trees provide an invaluable renewable resource that lends itself to multiple commercial applications and are therefore a major potential source of livelihood creation. In addition, the agro-forestry value chain has the rare distinction of being able to contribute to augmenting energy capacity and ecological wealth besides supporting the core needs of food and housing. As the primary resource for paper based industry, it supports the spread of education, so crucial for developing economies like ours. With increasing population, rising incomes and consequent demand for energy, education and housing including home lifestyle products, the wood-based value chain is well poised not only as a profitable economic activity but also as a catalyst sector that promotes sustainability and inclusive growth.

Forestry represents the second-largest land use in India after agriculture, but contributes merely 1% of GDP. This points to the dormant potential of this sector. China's experience also shows what can be unleashed by developing a competitive wood-based industry. According to a Study by the Center for International Forestry Research (CIFOR) and others, Chinese exports of Forest Products, in a span of just 8 years preceding 2005, rose from US \$ 3.6 billion to US \$ 17.2 billion, and in specific products like plywood, the increase was as much as ten-fold, making China one of the world's largest exporters. The United States and European Union were the main buyers with imports increasing 700 to 900%.

A concerted effort to promote wood plantations in degraded wastelands in India could potentially change the fortunes of small and marginal farmers and consequently the opportunities in Rural India. Degraded wastelands in the country are estimated at 60 million hectares by the Planning Commission, out of which 33 million hectares constitute degraded forest lands. By converting these wastelands into productive and renewable agro-forestry resources, strong economic multipliers can be unleashed to significantly mitigate the challenges of livelihood, energy and environmental security. Let me briefly outline some of these multiple possibilities.

◆ As a source of renewable and carbon-neutral biomass energy, agro-forestry can generate 1000 MW of power from a plantation of half a million hectares. This has the potential to substantially reduce the dependence on carbon generating fossil-fuel based energy. Specially suited for distributed energy systems in rural India, power generated from biomass can either meet the needs of smaller rural communities or be fed into the grid. Renewable plantations in 10 million hectares can produce as much as 20,000 MW of power, almost one-fourth of the total capacity addition proposed in the 11th Five Year Plan from conventional sources.

◆ Given the major environmental advantage of wood as a carbon sink, greater usage of wood in housing is a better alternative than the use of cement and steel which have much larger ecological footprints. By 2010-12, more than 500 million sq. ft. of space is expected to be built in India, mostly in urban areas. Even a limited use of wood for door and window frames, flooring, wooden cabinets etc would mean a demand for wood products valued at US \$ 3 billion. With greater usage of wood as a predominant material for housing, in rural and urban areas, this potential can increase exponentially.

◆ The wood-based industry is growing rapidly with increasing demand for furniture, housing construction material, packaging, agricultural implements, sports goods, plywood, veneer, matches, and so on. The Indian furniture market is estimated at US \$ 8 billion, and today employs around 3,00,000 people. Wooden home furniture is the largest segment. Imports of furniture are growing at a fast clip with a rate as high as 60%. Total wood-based import for various uses today is already over US \$ 1 billion. Depending on the regeneration of wood and the changing demand profile, India is likely to face a wood deficit of 20 to 70 million cubic metres by 2020. Certainly, the case for agro-forestry is obvious.

◆ Trees are also a valuable source of bio-chemicals. The cellulose part of the structure, used primarily for pulp and paper, represents only 20 to 30% of the total, and the balance can be a valuable resource for extraction of bio-chemicals with wide applications from food to medicine. Non-Timber

